

CS104 Project- Howzatt! Cricket Scorekeeper

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Chapter 1

Introduction

1.1 Instructions

The Howzatt! Cricket Scorekeeper is a web application which provides real-time scoring for cricket matches. It allows user to

- Team selection from predefined IPL teams (RCB, MI, CSK, KKR, DC)
- Dynamic score updating with all standard cricket events
- Detailed scorecards for both innings
- Match summary with winner declaration

Chapter 2

Basic Parts

2.1 Data Management

```
let data={
  team1: null,team2: null,
  toss_winner: null,
  innings: 1,
  batting_team: null,
  bowling_team: null,
  overs:0,
  noball: false
}

let player_data={
}
```

```
let score_data={
  runs:0,
  wickets:0,
  overs:0,
  balls:0,
  current_striker:{ ...
},
  current_non_striker:{ ...
},
  inning1_score: { ...
    },
  inning2_score: { ...
    wickets: 0,  },
  current_bowler: { ...
},
  crr: 0.0
}
```

The following objects are defined in score.js for storing information of the match

- scoredata(): tracks runs, wickets etc., current striker, non-striker, bowler etc.

- `playerdata()`: tracks data of all players of both the teams
- `data()`: stores data for bowling, batting team, etc.

2.1.1 Data Persistence

The application uses `localStorage` to persist data across pages:

1. User inputs match details
2. Data is stored in `localStorage`
3. Pages read from `localStorage` (using `getItem()`)
4. Update `localStorage` as required (using `setItem()`)

```
localStorage.setItem("data", JSON.stringify(match_data))
localStorage.setItem("player_data", JSON.stringify(player_data))
localStorage.setItem("score_data", JSON.stringify(score_data));
```

```
match_data = JSON.parse(localStorage.getItem("data")); //extracts
score_data = JSON.parse(localStorage.getItem("score_data")) //cant
player_data = JSON.parse(localStorage.getItem("player_data"))
```

2.2 Key Features

2.2.1 Setup Page

MATCH SETUP

RCB	DC	CSK	MI	KKR
Faf du Plessis	David Warner	Ruturaj Gaikwad	Rohit Sharma	Rohitavulak Gutkez
Virat Kohli	Prithvi Shaw	Devon Conway	Zabir Hossain	Venketesh Iyer
Glen Maxwell	Michael Marsh	Ajinkya Rahane	Suryakumar Yadav	Nitish Rana
Abhishek Sharma	Rishabh Pant	Ravindra Jadeja	Sanjay Singh	Rohit Singh
Shikhar Dhawan	Abhinav Prasad	Shreyas Iyer	Andre Russell	Shardul Thakur
Wanindu Hasaranga	Lalit Yadav	R.S. Dhoni	Nahid Bhuiyan	Sant Narine
Harshad Patel	Avinash Murtiga	Deepak Chahar	Jasprit Bumrah	Lockie Ferguson
Mohammed Siraj	Kuldeep Yadav	Mahesh Thakur	Piyush Chawla	Unmesh Yadav
Zash Hazare	Kuldeep Yadav	Tushar Deshpande	Zaheer Khan	Venugopal Reddy
Karn Sharma	Rohit Sharma	Mohammed Siraj	Kumar Kartikeya	Harshit Rana

Select Team 1

CSK

Select Team 2

MI

Toss Winner:

TEAM 1

Toss Decision:

bat

Enter Number Of Overs:

Number Of Overs

Show Result

- Dynamic team selection dropdowns
- Toss simulation
- Overs configuration
- Initial player selection modal (Details in custom)

2.2.2 Live Scoring

<

- Real-time updation: using updatescore() function wherein getElement-ById() option is used
- Automatic strike rotation
- Over progression tracking
- Wicket handling
- New batter and bowler selection (Details in custom)

2.2.3 Scorecards

Handled using previously defined object variables and boolean variables

Scorecard Page						
Go back to live						
Innings 1 Innings 2						
S.No.	Name	Runs	Balls Faced	4s	6s	Strike Rate
1	Rahit Sharma	21	6	1	0	350.00
2	Ishan Kishan	48	10	0	5	480.00
3	Tim David	3	1	0	0	300.00
S.No.	Name	Overs	Maidens	Runs Conceded	Wickets	Economy Rate
1	Rohitash Baskwad	2.0	0	39	1	3.25
2	Ravindra Jadeja	1.0	0	33	0	5.50
3	M.S Dhoni	0.0	0	0	0	0.00

- Batting and bowling statistics: shows only the stats of those players where a certain boolean value, eg. bowled or batted is true
- Player performance metrics

Chapter 3

Custom Features

3.1 Advanced Scoring

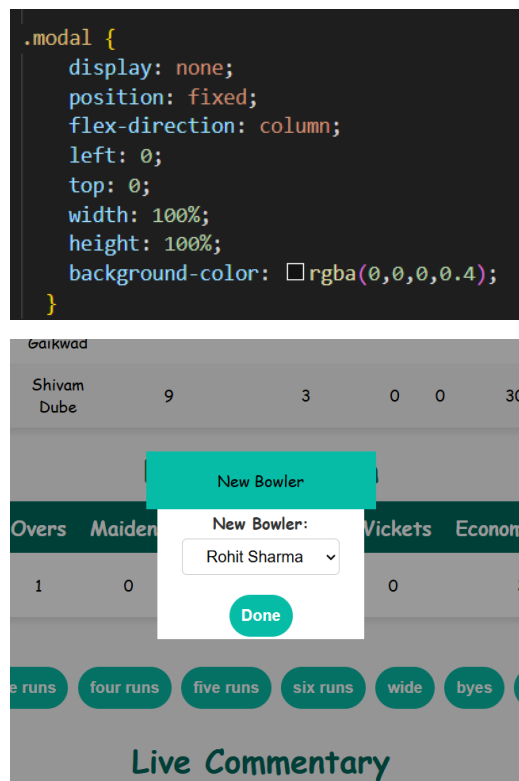
Implemented all cricket extras:

- Wides (+1 run, no ball counted): Implemented in the same way normal functions were implemented
- No Balls (free hit next delivery)
- Byes/Leg Byes (runs not credited to batter)
- Run outs (with runs completed) (Implemented using Modal)

3.2 User Interface

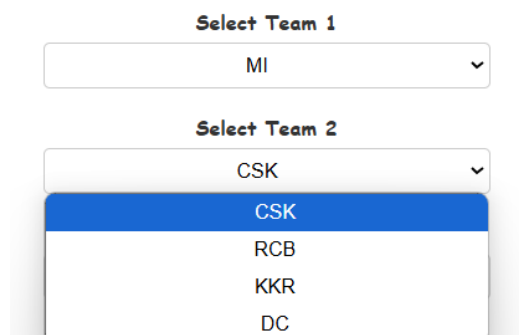
3.2.1 Modals

A modal is a dialog box/popup window that is displayed on top of the current page:



3.2.2 Interactive Elements

OnChange option is used in Event Listener. Now, the required data is stored using OnClick function in the dialog box. It calls a function when a user changes the selected option of a `<select>` element.



```

const team1=document.getElementById("team1_name")
const team2=document.getElementById("team2_name")
team1.addEventListener('change', ()=> {
  teamss = ['CSK', 'MI', 'RCB', 'KKR', 'DC']
  const selected_value = team1.value;
  team2.innerHTML = ""
  for (let i = 0; i < teamss.length; i++) {

    if (teamss[i] != selected_value) {

      const option_element=document.createElement("option");
      option_element.value=teamss[i]
      option_element.text=teamss[i]
      team2.appendChild(option_element)
    }
  }
})

```

Using Modal Box and OnChange Events

- Dynamic player dropdowns that
- Responsive scoring buttons
- Real-time commentary feed

Chapter 4

Conclusion

The Howzatt! Cricket Scorekeeper successfully implements:

- Complete cricket scoring rules
- Real-time match tracking
- Data persistence across pages
- Intuitive user interface

Bibliography

- [1] *Understanding Local Storage*, <https://blog.logrocket.com/localstorage-javascript-complete-guide/>
- [2] *Understanding HTML*, <https://www.w3schools.com/Html/>
- [3] *Understanding Styling*, <https://www.w3schools.com/css/>
- [4] *Understanding Logic*, <https://www.w3schools.com/js/>
- [5] *Creating A Modal Box*, https://www.w3schools.com/howto/howto_css_modals.asp
- [6] *Onchange Event* https://www.w3schools.com/jsref/event_onchange.asp